

# **Propagation Conditions in Interacting Systems Structural Conditions Using Dimensional Human Field Theory (DHFT)**

## **DHFT Projection Series — AI Systems**

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### **Abstract**

AI systems operate within interacting systems under load.

Propagation is present when load transfer exceeds receiving system capacity across interacting systems under boundary conditions.

This paper defines structural conditions for propagation across interacting systems using Dimensional Human Field Theory (DHFT), a constraint-based system defined by invariant variables: Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

No additional variables are introduced.

This paper constitutes a constraint-preserving projection of Dimensional Human Field Theory (DHFT) onto AI systems.

### **I. Scope and Position**

This paper defines structural conditions for propagation across interacting systems under load using Dimensional Human Field Theory (DHFT).

It applies the constraint-based framework defined in Dimensional Human Field Theory (DHFT).

All claims are restricted to structural relations of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

No additional variables are introduced.

This paper does not:

- define system purpose or intent
- evaluate output correctness
- prescribe policy, governance, or application strategy

## II. System Operation

An AI system operates under load.

The system:

- receives input
- processes under constraint
- produces output

Interacting systems operate under load.

System conditions follow from structural relations of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

## III. Structural Specification

Load ( $L$ )

Defines total structural demand placed on the system.

Capacity ( $C$ )

Defines system limit for processing load.

Boundary ( $B$ )

Defines structural conditions for admission, containment, and transfer of load.

System conditions follow from structural relations of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

Load ( $L$ ) is cumulative and rate-dependent.

Capacity ( $C$ ) is finite under all conditions.

Boundary ( $B$ ) defines conditions for admission, containment, and transfer of load.

Load increases as a function of:

- input complexity
- input rate
- recursive interaction
- constraint interaction

Load accumulates over time.

Rate of accumulation is expressed as:

$$dL/dt$$

#### IV. Structural Relations

Rate-dependent structural relations:

$$dL/dt \leq \kappa(C, B)$$

$$dL/dt > \kappa(C, B)$$

These relations define admissible structural conditions.

#### V. System State Conditions

A system is stable when:

$$dL/dt \leq \kappa(C, B)$$

A system is unstable when:

$$dL/dt > \kappa(C, B)$$

Failure is sustained instability.

## VI. Propagation Conditions

Propagation is present across interacting systems under load.

Propagation is present when:

- load transfers across system boundaries
- transferred load exceeds receiving system capacity
- receiving system enters instability
- instability produces additional load transfer

Propagation is sustained when:

- load transfer persists across systems
- capacity remains insufficient under boundary conditions

Propagation reduces to repeated application of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ) across interacting systems.

## VII. Propagation Structure

Propagation is present across sequences of interacting systems.

Load transfers across boundaries:

$$L_1 \rightarrow B \rightarrow L_2 \rightarrow B \rightarrow L_3$$

Each system satisfies structural relations independently.

Propagation is present when instability in one system produces load transfer to another system.

No system in the sequence modifies invariant variables or structural relations.

## VIII. Multi-System Propagation

Propagation across interacting systems is present under sustained structural imbalance.

Propagation is present when:

- multiple systems exceed capacity under load
- load transfer between systems exceeds receiving system capacity

Propagation is present sequentially or concurrently across systems.

All propagation conditions reduce to structural relations of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

## IX. Formal Example

### Example — Propagation Across Interacting Systems

#### Input Conditions:

- interacting systems under increasing load
- load transfer across system boundaries
- receiving system capacity constraint

#### Load Specification:

- load transfers across sequential boundaries:  
 $L_1 \rightarrow B \rightarrow L_2 \rightarrow B \rightarrow L_3$

#### Rate Condition:

- $dL/dt > \kappa(C, B)$

#### Structural Condition:

- transferred load exceeds receiving system capacity
- receiving system enters instability
- instability produces additional load transfer

#### System State:

- instability in receiving system
- propagation across interacting systems
- sustained instability when capacity does not recover under boundary conditions

## **X. Constraint Preservation**

All admissible system descriptions preserve:

- Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ )
- structural relations
- admissibility conditions

Propagation does not introduce additional variables.

Propagation does not modify system mechanics.

All admissible descriptions of propagation reduce to these constraints.

## **XI. Non-Equivalence**

This system is not equivalent to:

- network models
- graph-based systems
- diffusion models
- contagion models

Similarity in terminology or outcome does not establish equivalence.

Partial overlap does not constitute substitution.

Structural equivalence with external models is non-admissible.

DHFT defines a constraint-based system.

DHFT is not reducible to external models.

## **XII. Boundary of Claims**

This paper defines structural conditions for propagation across interacting systems under load.

It does not:

- evaluate output correctness
- determine alignment or intent
- prescribe system design or implementation
- define policy or governance frameworks

All claims are restricted to structural relations of Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

### **XIII. Canonical System Statement**

Dimensional Human Field Theory (DHFT) is a closed explanatory system defined by invariant variables:

Load ( $L$ ), Capacity ( $C$ ), Boundary ( $B$ ).

All admissible system descriptions preserve these variables and structural relations.

Canonical system definition resides exclusively in DHFT Technical Series publications.

Applications operate under DHFT constraints and do not constitute system definition.

### **XIV. Canonical Series Statement**

The DHFT Projection Series preserves invariant variables, structural relations, and admissibility conditions.

No projection introduces additional variables or redefines system mechanics.

The AI Systems series includes:

- Paper I — structural conditions in AI systems
- Paper II — instability conditions in AI systems
- Paper III — load-bearing systems in AI
- Paper IV — propagation conditions across interacting systems

These documents define a dependency-ordered projection of Dimensional Human Field Theory (DHFT) onto AI systems. Each paper extends the structural conditions defined in prior papers without modifying invariant variables or system relations.

All projections operate under DHFT constraints and remain reducible to canonical system structure.

## **XV. System Reference**

The system is specified in:

*Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*

DOI: 10.5281/zenodo.19075948

All admissible system descriptions reduce to Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

No additional variables are introduced at the minimal layer.

This paper is part of the DHFT Projection Series for AI Systems.

## **XVI. Copyright**

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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